

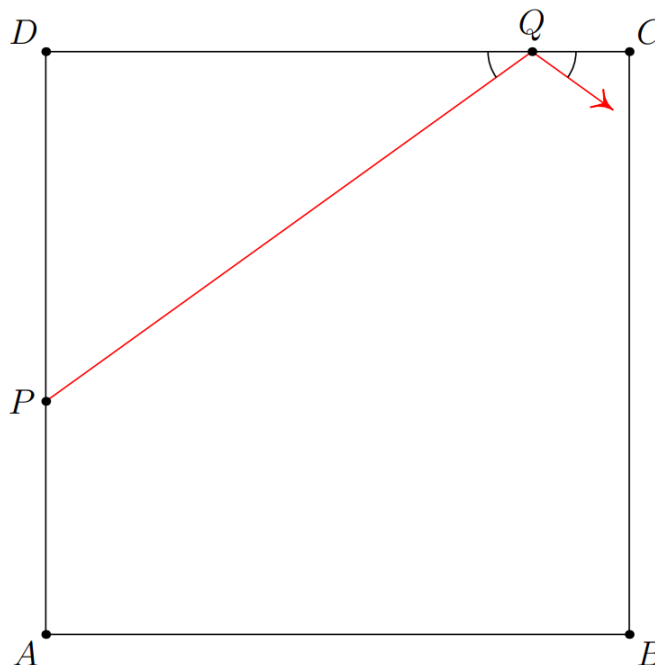
2025 AMC 10B Problem 25 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 25. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 25

Problem:

Square $ABCD$ has sides of length 4. Points P and Q lie on $\overline{AD}$ and $\overline{CD}$, respectively, with $AP = \frac{8}{5}$ and $DQ = \frac{10}{3}$. A path begins along the segment from P to Q and continues by reflecting against the sides of $ABCD$ (with congruent incoming and outgoing angles). If the path hits a vertex of the square, it terminates there; otherwise it continues forever. At which vertex does the path terminate?



Answer Choices:

- A. A
- B. B
- C. C
- D. D
- E. The path continues forever

 Join the Discussion

Stuck on this problem or want to share your approach?

Continue the conversation and see what others are thinking: [View Forum Thread](#) 

Solution:

The square $ABCD$ has a side length 4 and is placed in the coordinate plane as

$$A = (0, 0), \quad B = (4, 0), \quad C = (4, 4), \quad D = (0, 4).$$

When the path reflects off the sides of the square at equal angles, we can unfold the motion:

- ▶ Reflect the square repeatedly across its sides, tiling the plane with congruent 4×4 squares.
- ▶ In this unfolded tiling, the trajectory is a straight line passing through P and Q .

The path hits a vertex of the original square precisely when the line (l) passes through the lattice points $(4p, 4q)$. We will first find the equation of the line through PQ . Point P lies on $\overline{AD}$ with $AP = \frac{8}{5}$. Since $A = (0, 0)$ and $D = (0, 4)$,

$$P = \left(0, \frac{8}{5}\right).$$

Point Q lies on $\overline{CD}$ with $DQ = \frac{10}{3}$. Since $D = (0, 4)$ and $C = (4, 4)$,

$$Q = \left(\frac{10}{3}, 4 \right).$$

Thus, the line through $P(0, \frac{8}{5})$ and $Q(\frac{10}{3}, 4)$ is

$$25y - 18x = 40. \tag{1}$$

Since we want the points to pass through $4p$ and $4q$, let's assume $x = 4p$ and $y = 4q$.

From (1), we have:

$$25y - 18x = 40 \implies 25q - 18p = 10 \tag{2}$$

From the above, it is clear that:

$$q = 2t, \quad t \in \mathbb{Z} \text{ and } p = 5s, \quad s, t \in \mathbb{Z}.$$

Substituting into (2), we get:

$$50t - 90s = 10 \implies 5t - 9s = 1. \tag{4}$$

So we have reduced the problem to solving the simpler Diophantine equation (4).

Now, we solve for the smallest values of s and t from $5t - 9s = 1$. The smallest positive value of $s = 1$, which works when $t = 2$ gives us $x = 20, y = 16$. At this point, we can draw a quick diagram shown below and find that the vertex is **(B) B**.

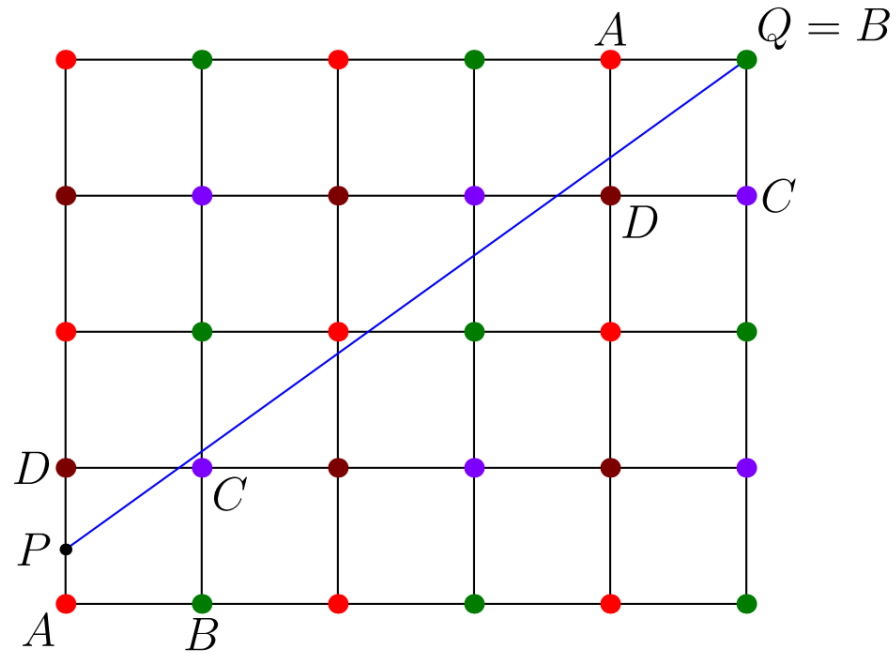
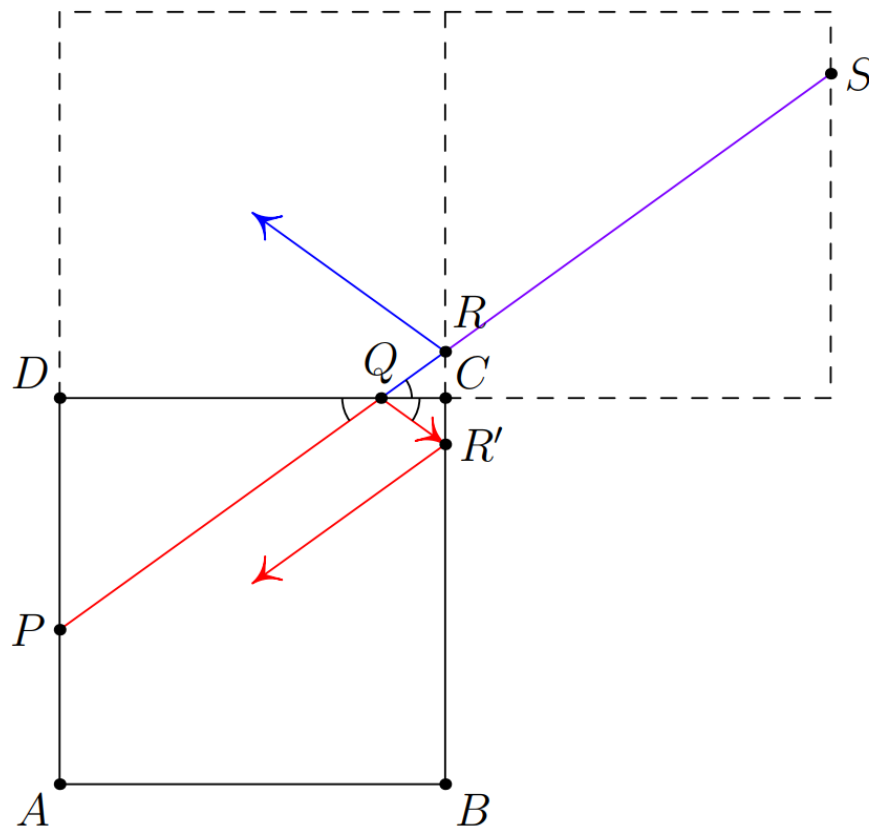


Figure 1: The above diagram shows the unfolded

Remark. If you haven't seen this concept before, this might be something new. Let's work through an example of unfolding these manually.



We can take a look at the diagram above, and the first few reflections. As the beam reflects at Q , it becomes very hard to track it. However, what if we pretended that the wall CD never existed, and that the beam could keep going?

At that point we reach the point R , where as our beam (red) reached the point R' . Note that QR and QR' are reflections in CD , as $\angle RQC = \angle R'QC$. This means that to find R' (and the path of the beam), we can instead follow the straight beam into the blue area, find R , and then reflect it to find R' .

We can keep going. As the straight beam with no walls enters the purple segment, we see that the blue ray is a reflection of the purple segment by the same logic. But since the blue arrow from R is a reflection of the red arrow from R' , we are simply chasing reflections.

By continuously chasing reflections in this manner we obtain our original image above, and can then find the pattern between vertices to find where it hits a vertex.

Remark. We can extract the answer more rigorously, as follows: each pair (p, q) describes how many squares we move by reflection, both horizontally and vertically. The parity of (p, q) determines which vertex of the original square is reached:

$p \bmod 2$	$q \bmod 2$	Vertex type
0	0	$A = (0, 0)$
1	0	$B = (4, 0)$
1	1	$C = (4, 4)$
0	1	$D = (0, 4)$

For our first hit,

$$p = 5 \equiv 1 \pmod{2}, \quad q = 4 \equiv 0 \pmod{2},$$

so the corresponding vertex in the original square is $B = (4, 0)$.

Remark. We can also find generic solutions to the above Diophantine equation using the method below:
From (4),

$$5t - 9s = 1.$$

Work modulo 9:

$$5t \equiv 1 \pmod{9} \implies t \equiv 2 \pmod{9}.$$

Therefore, we can Write t as

$$t = 2 + 9k, \quad k \in \mathbb{Z}.$$

Plug into (4) we obtain:

$$5(2 + 9k) - 9s = 1 \implies s = 1 + 5k.$$

Therefore

$$t = 2 + 9k, \quad s = 1 + 5k, \quad k \in \mathbb{Z}.$$

Recall

$$q = 2t = 2(2 + 9k) = 4 + 18k, \quad p = 5s = 5(1 + 5k) = 5 + 25k.$$

We are moving forward along the ray from P through Q , so we want the smallest nonnegative k giving a positive p, q . Taking $k = 0$,

$$p = 5, \quad q = 4.$$

Thus, the first vertex of the tiling hit by the line is

$$(x, y) = (4p, 4q) = (4 \cdot 5, 4 \cdot 4) = (20, 16).$$

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗

Powered by [Wiki.js](#)